Storage Estimation: Size × Count × Retention × Replicas

the factor of three most quick answers miss

Watch the 4-min video

In one line: storage = object size × new objects per day × days retained × replication factor, plus metadata, indexes and derived copies, adjusted for growth; the replication factor alone is usually 3×.

1 The formula

Users upload **500 million photos a day**, each about **2 MB**. How much storage do you need over five years? The quick answer misses a factor of three or more, and that is the gap between a sensible budget and a very unpleasant surprise.

```
storage = size × objects/day × days × replicas + overhead
```

Write each factor down, round it to an easy number, and multiply step by step. Saying each step out loud lets the interviewer follow and catch a slip early.



Figure 1. Four factors multiply into raw storage; overhead comes on top.

2 Why multiply by the replication factor

Every byte is stored more than once, so a failed disk or node never loses data.

- **Databases** keep a primary and replicas: one primary plus two replicas is 3 copies.
- **Distributed stores** such as Cassandra or HDFS commonly keep **3 copies**.
- **Object stores** like S3 replicate internally; you don't see the copies, but you pay for that durability in the price per GB.
- **Backups and cross-region copies** come on top, so the real number grows further.

3 Don't forget overhead

Extra	Typical size	Why it exists
Metadata row	~500 B to a few KB per object	owner, timestamp, location, tags
Indexes on metadata	+50–100% of the table size	fast lookups by owner, time, tag
Thumbnails, resized versions	+20–50% per photo	different screens and previews
Backups, other regions	+1 copy or more each	disaster recovery

4 Worked example: photos over five years

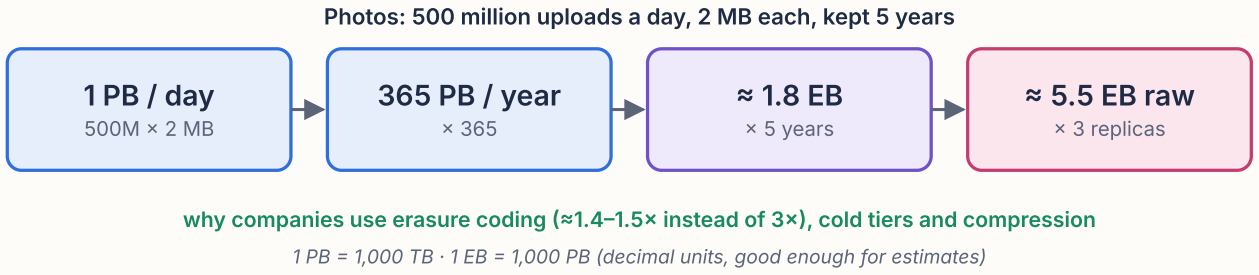


Figure 2. Multiply one factor at a time and keep the unit next to every number.

WORKED EXAMPLE

500M × 2 MB = 1,000 TB = **1 PB per day**. × 365 ≈ **365 PB per year**. × 5 years ≈ **1.8 EB**. × 3 replicas ≈ **5.5 EB of raw storage**.

Metadata is small next to that: 500M rows × 1 KB = 0.5 TB a day, about 0.9 PB over 5 years before indexes and replicas. But thumbnails at +20–50% add another 1–3 EB of raw storage. Exabytes are huge, which is exactly why companies use **erasure coding** (for example 10 data + 4 parity pieces, 1.4× overhead instead of 3×), **cold storage tiers** and **compression**.

5 Interview question: a chat app

Five-year storage for a chat app? State the assumptions first: **50M daily users**, each sending **40 messages a day**, about 100 bytes of text plus 100 bytes of metadata per message.

- Messages per day: 50M × 40 = **2 billion**.
- Per day: 2B × 200 B = **400 GB**.
- Five years: 400 GB × 1,825 days ≈ **730 TB**.
- × 3 replicas ≈ 2.2 PB, plus indexes: about **2.5–3 PB**.

Media attachments (photos, voice notes, video) go to object storage and are estimated separately with the same formula. They usually dwarf the text.

6 Plan for growth, then look for savings

Data rarely grows linearly. If users double every year, the last year writes as much as all earlier years together, plus a bit more. A flat “year 1 × 5” estimate can be several times too low.

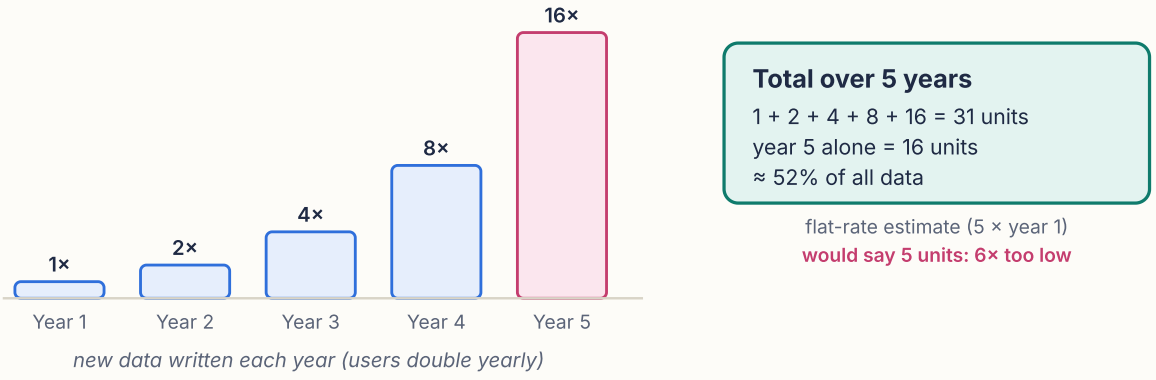


Figure 3. With yearly doubling, the fifth year alone is about half of the five-year total.

Savings: compress text (chat logs and JSON often shrink several times), move old data to cheaper tiers, expire data you don't need (TTLs, retention policies), and deduplicate identical uploads by content hash.

COMMON MISTAKES

- Forgetting the replication factor: a 3× underestimate.
- Ignoring metadata, indexes and thumbnails.
- Mixing bits and bytes, or GB and GiB, mid-calculation; pick decimal units and stay consistent.
- Assuming flat growth for a product that is doubling.
- Storing media in the database instead of object storage.

7 Key terms

Retention: how long data is kept

Erasure coding: data + parity pieces, ~1.4–1.5×

PB / EB: 10^{15} / 10^{18} bytes

Replication factor: number of copies, often 3

Cold tier: cheap, slow storage for old data

Derived copy: thumbnail, resized or transcoded version

Overhead: metadata, indexes, derived copies

Deduplication: store identical content once

Object storage: S3-style store for large blobs

8 Interview questions

Q1. How do you estimate 5-year storage? BASIC

Object size × new objects per day × days retained × replication factor, then add metadata, indexes and derived copies, and adjust for growth. Round each factor and multiply step by step.

Q2. Why multiply by the replication factor? BASIC

Each byte is physically stored several times, usually 3, for durability and availability, and you pay for every copy. Skipping it underestimates by 3×.

Q3. 500M photos a day at 2 MB, kept 5 years. Raw storage? SCENARIO

1 PB/day × 365 ≈ 365 PB/year × 5 ≈ 1.8 EB × 3 replicas ≈ 5.5 EB, before thumbnails and metadata.

Q4. Estimate 5-year storage for a chat app with 50M DAU. SCENARIO

40 messages each = 2B/day × 200 B = 400 GB/day × 1,825 ≈ 730 TB × 3 ≈ 2.2 PB, plus indexes ≈ 2.5–3 PB. Media goes to object storage, estimated separately.

Q5. Your estimate is 5 EB. How would you bring the cost down? SCENARIO

Erasure coding instead of 3 full copies (≈1.4× instead of 3×), cold tiers for old photos, compression, deduplication by content hash, and deleting unused derived sizes.

Q6. Users double every year. How does that change a 5-year estimate? TRICKY

Yearly new data goes 1, 2, 4, 8, 16: a total of 31 units, not 5. The final year is about half the total, so size for the last year, not the first.

Q7. Does S3 mean you can ignore replication? TRICKY

You don't multiply the bytes yourself, since S3 replicates internally and charges per logical GB. But you still pay for it in the price, and cross-region replication or backups add real copies you do pay for.

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